| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 12/27/2011 |
| **PI Name:** | Fornace, Albert M.D. | | |
| **Project Title:** | Mouse models approach for intestinal tumorigenesis estimates by space radiation | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | **TechPort:** | | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | af294@georgetown.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 202 687-7843 |
| **Organization Name:** | Georgetown University | | |
| **PI Address 1:** | Dept. of Oncology, Lombardi Comprehensive Cancer Center | | |
| **PI Address 2:** | Research Building, Room E504, 3970 Reservoir Rd., NW | | |
| **PI Web Page:** | | | |
| **City:** | Washington | **State:** | DC |
| **Zip Code:** | 20007-2126 | **Congressional District:** | 1 |
| **Comments:** | http://www9.georgetown.edu/ | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2006 Radiation Biology NNJ06ZSA001N |
| **Start Date:** | 04/01/2007 | **End Date:** | 09/30/2011 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Cucinott1a, Francis | **Contact Phone:** | 281-483-0968 |
| **Contact Email:** | noaccess@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received extension to 9/30/2011, per S. Krenek/JSC (01/2011)<br>NOTE: Start/end dates changed to 4/1/07-3/31/11, per update from P. Denkins/JSC; previously had 12/1/2006-11/30/2010 (9/2007) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX07AH70G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Ionizing radiation (IR) is a known risk factor for colorectal (CR) cancer. Considering the high spontaneous incidence of gastrointestinal (GI) cancer in the US and even higher incidence of pre-malignant lesions, such as colonic polyps, an even modest increase by radiation exposure could have a significant effect on health risk estimates for manned space flight. During space travel, astronauts are exposed to levels of radiation, which increases their risk for acute short-term deleterious effects, such as during a solar storm, and for long-term deleterious effects like cancer. Long duration space missions, like those at the International Space Station, lunar missions, or a mission to Mars, are of sufficient duration to allow for appreciable cumulative doses of cosmic radiation with potentially profound health consequences. This project is important because there are no good estimates for intestinal, particularly CR, cancer by space radiation. Considering the increased CR cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors and to investigate modulation of specific signaling pathways as a model for cancer prevention. We are employing APC mouse models that have been successfully used to demonstrate radiation-induced intestinal cancer. The APC models have inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, primarily adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. The strategy has been to compare results for space radiation, protons and HZE (56-Fe), on intestinal tumor development to that of low-LET (linear energy transfer) radiation (gamma rays). Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for humans exposed to space radiation. Many of the molecular events leading to intestinal tumors have been characterized. We will monitor and compare molecular changes in three types of radiation-induced tumors. The specific aims for this proposal are:<br>1. Quantitatively compare HZE ion or high-energy proton irradiation to gamma-ray effects on intestinal tumor incidence and grade using the Apc+/min tumor model.<br>2. Characterize and compare inactivation of wt APC locus on chromosome 18B1 after gamma vs. HZE ion or high-energy proton irradiation with the Apc+/min tumor model.<br>3. Determine intestinal tumor incidence in the Apc+/min; Ppm1d-/- mouse model and mechanisms for reactivation of the Apc+/min tumor phenotype by radiation.<br>4. Determine molecular signatures by comparative genomic hybridization (CGH) and expression profiling in spontaneous intestinal tumors compared to those induced by HZE ions, high-energy protons, and gamma irradiation.<br>5. Quantitatively compare HZE ion or high-energy proton irradiation vs. gamma-ray effects on incidence of other tumors. | | |
| **Rationale for HRP Directed Research:** | | | |
| **Research Impact/Earth Benefits:** | Gastrointestinal (GI) tumors are frequent in the U.S., and colorectal cancer (CRC) is the third most common cancer accounting for 10% of all cancer deaths. In 2008 there were an estimated 130,000 new cases of CRC in the U.S. with the frequency of precancerous lesions being present in about 10% of adults at age 40 (approximately the average age of a typical astronaut). The incidence of precancerous lesions increases to ~25% in older adults ( http://www.cdc.gov/ ). Considering the high frequency of CR tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. Long duration space missions, like those at the International Space Station as well as lunar and later Mars missions, increase astronauts' exposure to space radiation as well as the risks associated with such exposure. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and CR carcinogenesis. | | |

| | |
|---|---|
| **Task Progress:** | SUPPLEMENTAL REPORTING FOR FINAL REPORT--DECEMBER 2011<br>The overall plan is to apply the well-characterized APC mouse models (APC+/min and APC1638N) of intestinal tumorigenesis for development of risks estimates for various radiation exposures during manned space flight. To undertake a comparative study of intestinal tumor risk in mice following exposure to different types of radiation, the relative toxicity of 56-Fe particles and protons to gamma-rays needed to be determined so that we could then standardize with doses equitoxic to 2 and 5 Gy of gamma-radiation in C57Bl/6J mice; these gamma-ray doses have been shown to be effective in increasing intestinal tumorigenesis. The rationale for this initial study is that acute toxicity for the planned radiation sources in mice is not available. Proton and 56-Fe irradiation was performed at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. Results to date indicate that the relative biological effectiveness (RBE) for 56-Fe particles compared to gamma-rays is 1.25 and RBE for proton is 1.06. Although the RBE for protons was expected, the RBE for 56-Fe particles is surprisingly low and is an important parameter that has allowed us to undertake tumorigenesis study in APC mutant models. Irradiated mice are being observed long-term, and specific tissues are also provided to various NASA investigators to complement their own projects.<br><br>Subsequently we irradiated 6 to 8 week old female APC+/min mice with iron particles (4 Gy of 1 GeV/n 56-Fe ) and protons (4.76 Gy of 1 GeV/n) which is equivalent to 5 Gy of gamma-radiation. Including gamma-rays a total of 100 mice was irradiated and have been monitored for intestinal tumorigenesis. Mice were observed for 100 days and at the end of the observation period mice were humanely euthanized as per Georgetown University Animal Care and Use Committee (GUACUC) approved protocol and tumors counted and tumor samples harvested for qualitative analysis. The APC+/min mice show frequent spontaneous tumors and IR typically triggers a modest GI tumor increase in adults compared to larger increases in newborn or younger mice, and we also found a modest but significant increase after gamma-rays. In contrast, the increase in tumor number with equitoxic doses of protons or Fe ions radiation was significantly greater. Interestingly, in the course of our analyses, we also noted a larger increase of clustered tumors defined here as five or more adenomas without intervening normal mucosa with space radiation compared to gamma rays. Results to date indicate the space radiation, both high-energy protons and 56-Fe particles, may have unique properties conferring an increased tumorigenic risk compared to gamma rays. Quantitative and qualitative study with 5 Gy gamma and equitoxic doses of proton and iron has been completed and provided us with meaningful data to develop risk estimates and countermeasure strategies.<br><br>We have also completed irradiating APC+/min mice with 2 Gy gamma and equitoxic doses of proton (1.88 Gy) and iron (1.6 Gy). Including gamma-rays and unirradiated control, a total of 120 mice was irradiated and were monitored for intestinal tumorigenesis. Mice were observed for 120 days and at the end of the observation period mice were humanely euthanized. Increase in tumor number with equitoxic doses of protons or Fe ions radiation was significantly greater than gamma radiation. Between 5 and 2 Gy tumorigenesis results we observed a radiation dose dependent increase in tumorigenesis.<br><br>For tumorigenesis study we also used another APC mutant model, APC1638N. We carried out exposures to 5 Gy of gamma rays and to equitoxic doses of 1 GeV/n protons and 1 GeV/n 56-Fe ions at NSRL. Female and male mice were exposed at the age of 6-8 weeks and animals were sacrificed 150 days after irradiation. We report here that in Apc1638N mice 5 Gy of gamma rays and equitoxic doses of space radiation induce the same increase on overall intestinal tumor multiplicity. However, when colon tumorigenesis was considered as an end-point, Fe ions induced the greatest effect on both tumor number and grade. The level of spontaneous tumors was markedly less with this model and has allowed us to extend our studies to less than 1 Gy of Fe or protons. Tissues have been archived and have been made available to multiple NASA-supported laboratories and this repository should be a valuable resource for the NASA life science program. Overall, our results indicate that the relative tumorigenic risk for space radiation is elevated compared to equal doses or equitoxic doses of gamma radiation. 56-Fe particles, in particular, appear to have unique properties compared to low LET radiation that confer long-term changes in intestinal epithelium that are relevant to cancer risk.<br><br>ANNUAL PROGRESS REPORT-- FEBRUARY 2011<br><br>The overall plan is to apply the well-characterized APC mouse models (APC+/min and APC1638N) of intestinal tumorigenesis for development of risks estimates for various radiation exposures during manned space flight. To undertake a comparative study of intestinal tumor risk in mice following exposure to different types of radiation, the relative toxicity of 56Fe particles and protons to gamma-rays needed to be determined so that we can standardize with doses equitoxic to 2 and 5 Gy of gamma-radiation in C57Bl/6J mice; these gamma-ray doses have been shown to be effective in increasing intestinal tumorigenesis. The rationale for this initial study is that acute toxicity for the planned radiation sources in mice is not available. Proton and 56-Fe irradiation was performed at the NSRL at Brookhaven National Laboratory. Results to date indicate that the relative biological effectiveness (RBE) for 56-Fe particles compared to gamma-rays is 1.25 and RBE for proton is 1.06. Although the RBE for protons was expected, the RBE for 56Fe particles is surprisingly low and is an important parameter that will allow us to plan long-term experiments. Irradiated mice are being observed long-term, and specific tissues are also being provided to various NASA investigators to complement their own projects.<br><br>Subsequently we irradiated 6 to 8 week old female APC+/min mice with iron particles (4 Gy of 1 GeV/n 56-Fe ) and protons (4.76 Gy of 1 GeV/n) which is equivalent to 5 Gy of gamma-radiation. Including gamma-rays a total of 100 mice was irradiated and have been monitored for intestinal tumorigenesis. Mice were observed for 100 days and at the end of the observation period mice are being humanely euthanized as per Georgetown University Animal Care and Use Committee (GUACUC) approved protocol and tumors counted and tumor samples harvested for qualitative analysis. The APC+/min mice show frequent spontaneous tumors and IR typically triggers a modest GI tumor increase in adults compared to larger increases in newborn or younger mice, and we also found a modest but significant increase after gamma-rays. In contrast, the increase in tumor number with equitoxic doses of protons or Fe ions radiation was significantly greater. Interestingly, in the course of our analyses, we also noted a larger increase of clustered tumors defined here as five or more adenomas without intervening normal mucosa with space radiation compared to gamma rays. Results to date indicate the space radiation, both high-energy protons and 56Fe particles, may have unique properties conferring an increased tumorigenic risk compared to gamma rays. Quantitative study with 5 Gy gamma and equitoxic doses of proton and iron has been completed and qualitative study to understand the mechanism of space radiation induced enhanced tumorigenesis is ongoing.<br><br>We have also completed irradiating APC+/min mice with 2 Gy gamma and equitoxic doses of proton (1.88 Gy) and iron (1.6 Gy). Including gamma-rays and unirradiated control, a total of 120 mice was irradiated and have been monitored for intestinal tumorigenesis. Mice were observed for 120 days and at the end of the observation period mice are being humanely euthanized. Increase in tumor number with equitoxic doses of protons or Fe ions radiation was significantly greater. Tumor samples harvested during quantitative study are now being processed for qualitative analysis.<br><br>For the APC1638N model we carried out exposures to 5 Gy of gamma rays at GU and at CERF, and to equitoxic doses of 1 GeV/n protons and 1 GeV/n 56-Fe ions at NSRL. Female and male mice were exposed at the age of 6-8 weeks and animals were sacrificed 150 days after irradiation. We report here that in Apc1638N mice 5 Gy of gamma rays and equitoxic doses of space radiation induce the same increase on overall intestinal tumor multiplicity. However, when colon tumorigenesis was considered as an end-point, Fe ions induced the greatest effect on both tumor number and grade.<br><br>Based on our most recent data, the APC1638N appears to be a promising model for in vivo estimates of space radiation-induced intestinal tumorigenesis. Also, by employing this model we will be able to investigate molecular mechanisms and effects on stem cells and their niche underlying space radiation-induced GI cancer progression.<br><br>In the course of year 4, we have also started our study on effects of 1 Gy of gamma rays and equitoxic doses of 1 GeV/n protons and 1 GeV/n 56-Fe ions. Data are already available for gamma irradiated animals: we observed that even with 1 Gy and lower doses the signal to noise ratio for colon tumors in APC1638N is significantly high, with irradiated animals developing multiple lesions, but not control animals. Furthermore, animals exposed to lower doses survive up to 9 months after exposure and the size of colonic tumors is significantly larger than in 5 Gy irradiated mice. Histopathology analysis of tumors from 1 Gy-irradiated animals is ongoing. |
| **Bibliography Type:** | Description: (Last Updated: 06/30/2026) |
| **Abstracts for Journals and Proceedings** | Trani D, Datta K, Tyburski J, Moon B-H, Fornace AJ Jr. "Acute and late intestinal effects induced by exposure to HZE and high-energy protons beyond low earth orbit (LEO): Mouse Model Approaches." 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010.<br>2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. Abstract #1138. http://www.dsls.usra.edu/meetings/hrp2010/pdf/Postdocs/1138Trani.pdf , Feb-2010 |
| **Abstracts for Journals and Proceedings** | Datta K, Trani D, Kallakury B, Cole M, Fornace AJ. "Accelerated hematopoetic toxicity due to ablation of myeloid progenitors: A hallmark of heavy ion radiation." Presented at the 56th Annual Meeting of Radiation Research Society, Maui, HI, September 25-29, 2010.<br>56th Annual Meeting, Radiation Research Society, September 25-29, 2010.<br>http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=e50705ed-1fdf-4831-b075-cdfd85dfa8b3&cKey=2d6e5f19-7477-4931-842a-19424f926ee8&mKey=5fb93787-2553-4b3b-a915-8c1ae71f127a , Sep-2010 |
| **Abstracts for Journals and Proceedings** | Trani D, Datta K, Moon BH, Sidahmed E, Sun J, Kallakury B, Fornace AJ Jr. "Enhanced intestinal tumor multiplicity and grade in vivo after HZE and proton exposure: mouse models for space radiation risk estimates." 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, September 25-29, 2010.<br>56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, September 25-29, 2010.<br>http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=9e011f35-da29-44e9-81ca-4b52cf617fb4&cKey=c905d5d2-7916-4fa1-b6a7-c7c517ad48f8&mKey=5fb93787-2553-4b3b-a915-8c1ae71f127a , Sep-2010 |
| **Abstracts for Journals and Proceedings** | Suman S, Datta K, Trani D, Fornace AJ. "Space radiation induced intestinal tumorigenesis is mediated via preferential activation of ß-catenin/TCF-4 signaling." 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011.<br>14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. Abstract Book, p. 160. , Sep-2011 |
| **Articles in Peer-reviewed Journals** | Datta K, Suman S, Trani D, Doiron K, Rotolo JA, Kallakury BV, Kolesnick R, Cole MF, Fornace AJ. "Accelerated hematopoietic toxicity by high energy (56)Fe radiation." Int J Radiat Biol. 2011 Dec 12. [Epub ahead of print] PubMed PMID: 22077279 , Dec-2011 |
| **Articles in Peer-reviewed Journals** | Trani D, Datta K, Doiron K, Kallakury B, Fornace AJ Jr. "Enhanced intestinal tumor multiplicity and grade in vivo after HZE exposure: mouse models for space radiation risk estimates." Radiat Environ Biophys. 2010 Aug;49(3):389-96. Epub 2010 May 20. PMID: 20490531 , Aug-2010 |